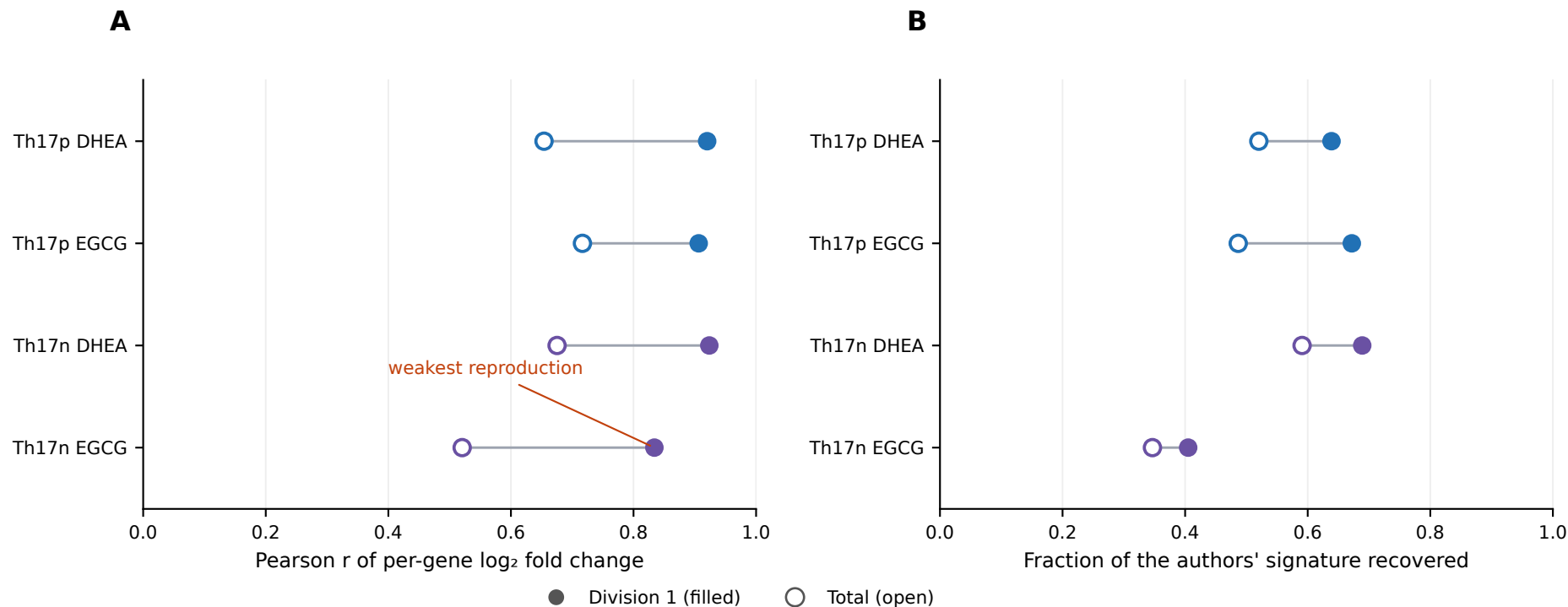


The published bulk contrasts reproduce on the first-division gate

GSE290297, 79 deposited TPM libraries | our moderated-t contrasts against the authors' Table S3, per cell type and inhibitor, computed separately in each division gate

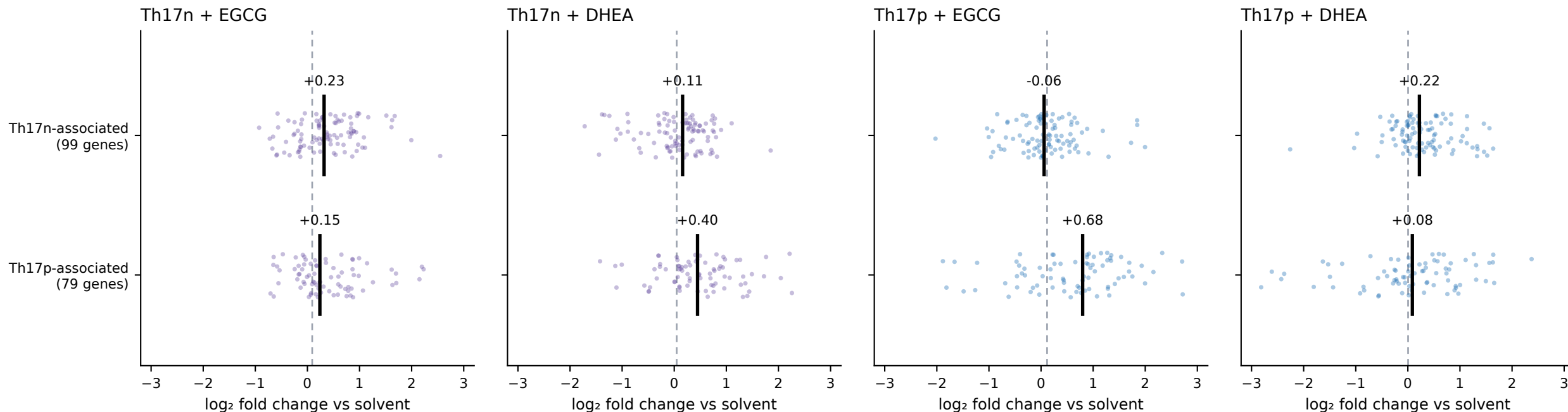


Agreement with Table S3 is reproduction of the authors' own analysis of their own deposit, not independent replication. The deposit declares no animal field, so every estimate is library-level. The gate is inferred from this agreement; the deposit does not state it.

In Th17n, PGAM inhibition moves both gene groups, not the pro-inflammatory group alone

GSE290297 division-1 libraries | genes partitioned by the deposit's own DMSO Th17p-versus-Th17n contrast (BH ≤ 0.05 and $|\log_2 \text{FC}| \geq 1.5$), then plotted against each inhibitor

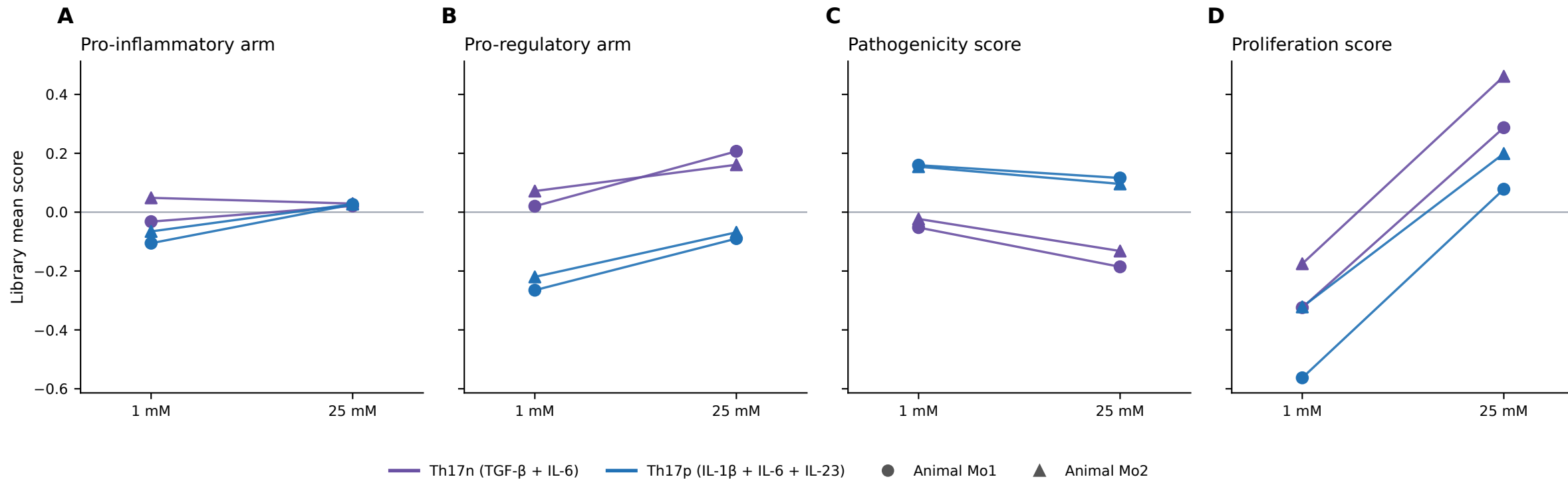
A Dashed line: median shift of the 11,003 genes in neither group, the whole-transcriptome move each inhibitor produces. Numbers are each group's median after removing it.



Library-level descriptive contrasts on deposited TPM; no animal field exists, so none of this is animal-level inference. Group medians are post-hoc readouts of the governed table. Centring removes the global shift but does not make the groups independent.

The pathogenicity score rises at low glucose through the pro-regulatory arm in 4 of 4 animal pairs

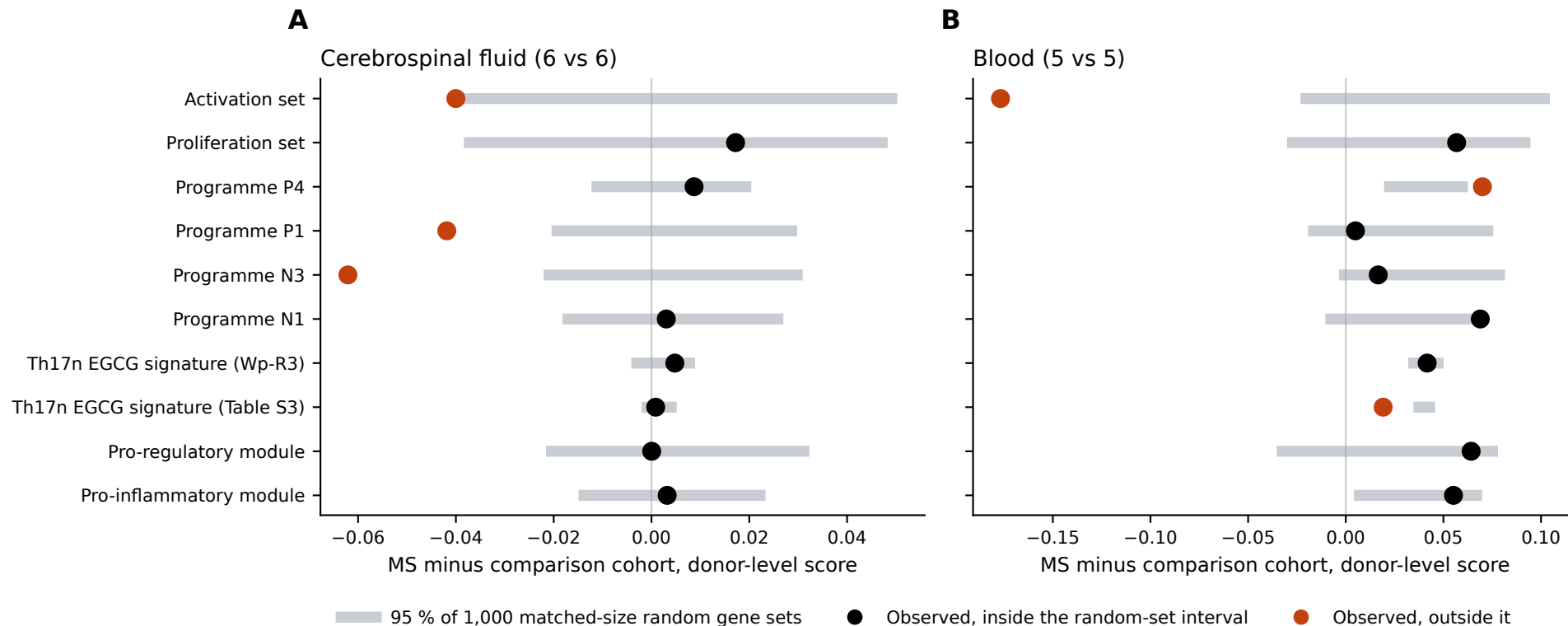
GSE289733, 8 libraries from 2 animals crossed over all four conditions | each line is one animal's paired libraries; scores use the authors' Table S1 modules with their highly-variable-gene flags



Two animals. Each panel shows four paired differences and no test is performed; cell-level standard errors would describe cells, not mice. Scores are RNA module summaries, not measurements of pathogenicity, enzyme activity or flux.

In blood, random gene sets separate the two human cohorts as well as the modules do

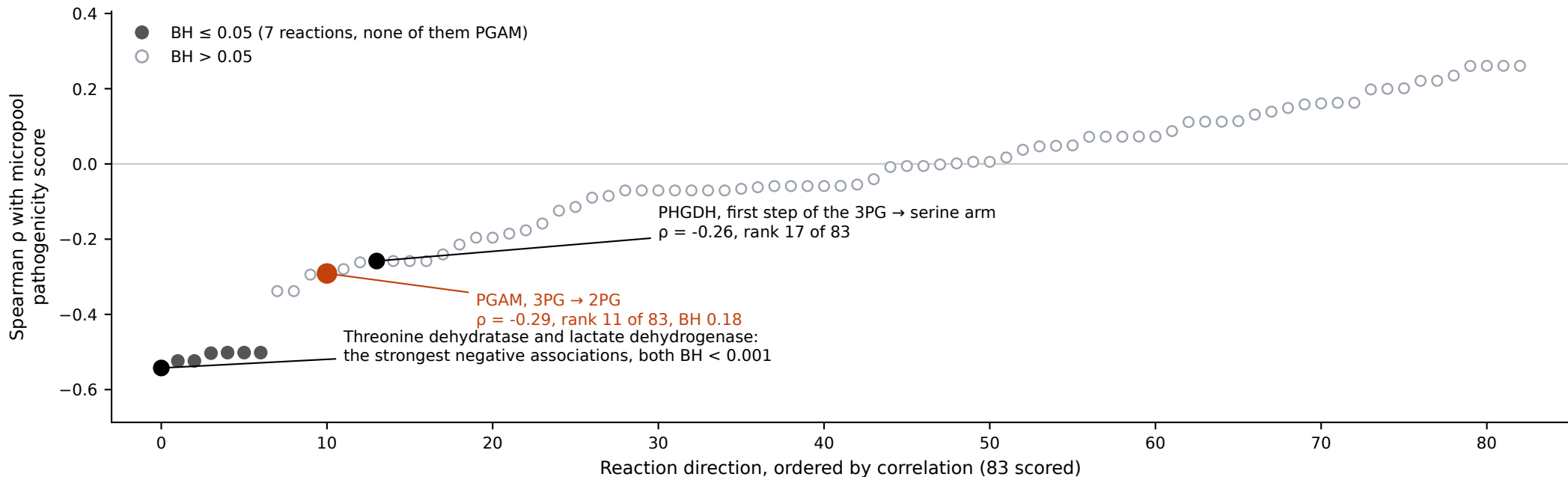
GSE138266 reused | 12 donors, 6 multiple sclerosis and 6 idiopathic intracranial hypertension;
35,928 CD4-lineage T cells; each random set is drawn at the same mapped size and scored identically.
The two panels have different x scales: the blood differences are about three times larger.



Donor-level transport of exposed mouse-derived signatures into a human dataset collected for another purpose; the comparison cohort is not healthy donors. A Mann-Whitney test on 5 versus 5 cannot fall below $p = 0.0079$. The pathogenicity score is a difference of two gene sets and has no matched random comparator, so it is not drawn. The random-set null was specified after the first run and is a post-hoc amendment.

PGAM keeps the published negative sign but is not the most distinctive reaction

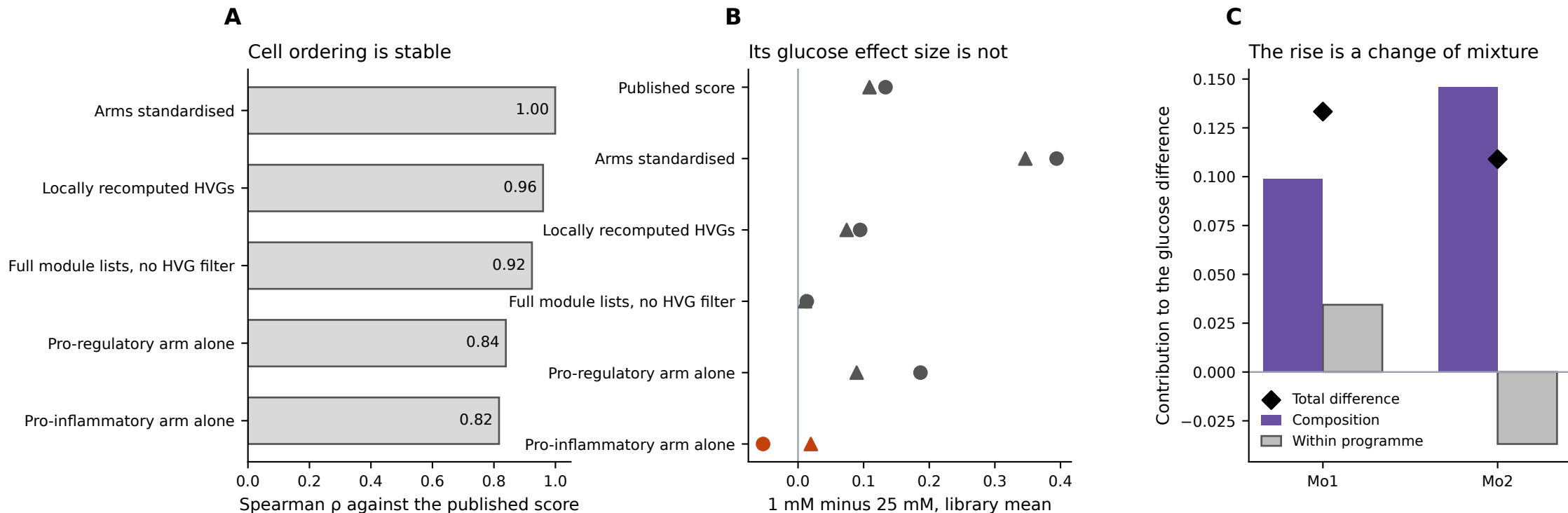
Compass 1.0.0 on 60 micropools of Wp-R1 Th17n cells | version-and-input sensitivity, not Figure 1:
the published scVI-imputed input was never deposited and the scope is two RECON2 subsystems



Pools are not independent replicates and share animals and cultures, so the nominal degrees of freedom overstate the information. Compass scores are model outputs from RNA, not flux. Restricting the reaction set puts the published ~ 900 -reaction ranking out of scope. Ranks are the governed run's own tie-aware values, so tied reactions share a rank.

The cell ranking survives how the score was built; the glucose effect size does not

GSE289733 Th17n cells | A-B, Wp-P01 variant scores against the published definition; C, Wp-P03 Kitagawa decomposition of the same glucose difference into mixture and within-programme parts



Every variant uses the same cells and the same expression matrix, so agreement between them is the absence of a gene-selection artefact, not independent evidence. In B, circles are animal Mo1 and triangles Mo2. In C the within-programme term has opposite signs in the two animals and is reported inconclusive; programmes are defined from the same expression as the score, so part of the mixture term is guaranteed by construction.